

# $t\bar{t}$ Samples — 2024 (Run 3)

**RunIII2024Summer24 NanoAODv15**

Dataset survey for the  $H+c \rightarrow WW$  analysis

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## Scope & method

Queried DAS on lxplus with `dasgoclient` :

```
dasgoclient -query="dataset dataset=/TTto*/RunIII2024Summer24NanoAODv15*/NANOAODSIM"  
dasgoclient -query="summary dataset=<ds>"          # nevents / nfiles / size
```

**Campaign:** `RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2`

**Caution:** a bare `/TT*/` wildcard returns **hundreds** of BSM samples ( `TTALPto2Mu` , `TTtoUEMu-LFV` , ...). All listings here are filtered to **SM  $t\bar{t}$**  production only.

Tier: NANOAODSIM. Several alternative NanoAOD flavours also exist per dataset

( `BTVNanoV15` , `JMENanoV15` , `FS` FastSim) — the plain `150X_..._v2` reco is the analysis default.

## Nominal $t\bar{t}$ — inclusive decay channels

POWHEG + Pythia8, TuneCP5, 13.6 TeV

| dataset      | events        | files        | size           |
|--------------|---------------|--------------|----------------|
| TTto2L2Nu    | 470,123,263   | 780          | 1.49 TB        |
| TTtoLNU2Q    | 484,475,057   | 790          | 1.55 TB        |
| TTto4Q       | 472,535,695   | 773          | 1.49 TB        |
| <b>total</b> | <b>1.43 B</b> | <b>2,343</b> | <b>4.53 TB</b> |

Full names follow the pattern:

```
/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoA0Dv15-150X_mcRun3_2024_realistic_v2-v3/NANOA0DSIM
```

**Note the version suffix differs:** 2L2Nu is -v3, the other two are -v2.

## HT-binned $t\bar{t}$ (MadGraph MLM)

TT-3Jets\_Bin-HT-\*\_TuneCP5\_13p6TeV\_madgraphMLM-pythia8

| HT bin (GeV) | events       | files        | size           |
|--------------|--------------|--------------|----------------|
| 100–400      | 155,239,759  | 1,730        | 0.62 TB        |
| 400–800      | 102,074,220  | 1,195        | 0.54 TB        |
| 800–1500     | 52,531,094   | 549          | 0.32 TB        |
| 1500–2500    | 33,843,305   | 613          | 0.23 TB        |
| > 2500       | 32,720,384   | 517          | 0.23 TB        |
| <b>total</b> | <b>376 M</b> | <b>4,604</b> | <b>1.95 TB</b> |

Useful if the analysis needs enhanced statistics at high hadronic activity — complementary to, **not stackable with**, the inclusive POWHEG set.

## $t\bar{t}$ + heavy flavour

| dataset              | events        | files      | size           |
|----------------------|---------------|------------|----------------|
| TT4B (madgraph)      | 9,898,300     | 34         | 0.043 TB       |
| TTBBto2L2Nu (powheg) | 12,499,196    | 203        | 0.045 TB       |
| TTBBtoLNU2Q (powheg) | 22,473,402    | 283        | 0.081 TB       |
| TTBBto4Q (powheg)    | 14,990,580    | 189        | 0.053 TB       |
| <b>total</b>         | <b>59.9 M</b> | <b>709</b> | <b>0.22 TB</b> |

$t\bar{t}b\bar{b}$  matters for **c-tagging mis-ID** studies — the dominant heavy-flavour background to a charm-tagged signal.

$t\bar{t} + V$  ( TTZ-ZtoQQ , TTW-WtoQQ , TTWW , TTWZ , TTZZ , TTZH , TTWH )

also exist in this campaign; small cross sections, listed in the companion note.

## Systematic-variation samples (all available)

**Top mass** — `TTto2L2Nu_Par-MT-*`

`166p5` , `169p5` , `171p5` ,  
`173p5` , `175p5` , `178p5`

6 points,  $\pm 5$  GeV around nominal

$h_{damp}$  — `Par-Hdamp-*`

`158` , `418`

(nominal  $\approx 243$ )

**Tune / colour reconnection**

`TuneCP5Up` , `TuneCP5Down`

`TuneCP5CR1` , `TuneCP5CR2`

**Alternative generators**

`amcatnloFXFX` (2-Jets)

`madgraphMLM` (3-Jets)

`sherpaMEPS` (4-Jets, 1NLO3LO)

The same `Par-MT` / `Par-Hdamp` / `Tune*` variations exist for `TTtoLNU2Q` as well  
— so the standard modelling-uncertainty suite is fully covered for 2024.

## Comparison to the 2022postEE set in use

|                    | 2022postEE (current)           | 2024 (this survey)        |
|--------------------|--------------------------------|---------------------------|
| campaign           | RunIII2022EE...NanoAODv12      | RunIII2024Summer24...v15  |
| nominal $t\bar{t}$ | TTto2L2Nu , TTtoLNu2Q , TTto4Q | same 3, <b>NanoAODv15</b> |
| inclusive events   | —                              | <b>1.43 B</b>             |
| HT-binned          | not used                       | <b>available</b> , 376 M  |
| $t\bar{t}b\bar{b}$ | not used                       | <b>available</b> , 60 M   |

**NanoAOD version changes v12 → v15.** Branch content and c-tagger columns differ; the `higgscharm` processor's expected branches must be checked before any 2024 processing.

## Caveats before using these

1. **NanoAODv15  $\neq$  v12.** Verify branch names ( `Jet_btagPNet*` , c-tagger WPs, `LHE*` ) against the processor. The negrw training features ( `lhe_*` , `genparton_*` ) must all still exist.
2. **Do not mix** inclusive POWHEG with HT-binned MadGraph without a stitching prescription — they overlap in phase space.
3. **Multiple NanoAOD flavours** per dataset ( `BTVNanoV15` , `JMENanoV15` , `FS` ).  
Pick one deliberately; `150X_mcRun3_2024_realistic_v2` is the standard reco.
4. **Luminosity / golden JSON** for 2024 must be sorted before any data comparison.  
`Run2024A-I` exist with both `PromptReco` and `2024CDEReprocessing` .
5. **Cross sections** for 13.6 TeV 2024 — use the TOP PAG recommendation, not the 2022 values.



## Summary

**Nominal  $t\bar{t}$ : 1.43 B events / 4.53 TB** across 3 decay channels

**+ 376 M** HT-binned · **+ 60 M  $t\bar{t}b\bar{b}$**

Full systematic suite available (mass,  $h_{damp}$ , tune, CR, alt. generators)

### Blocking question before use:

does the `higgscharm` processor read NanoAODv15 branches?

Dataset lists + DAS commands: `Projects/HToWW/2026-07-25-tt-2024-samples/`